

6.2 Classwork: Quadratics — Markscheme

1. The diagram below shows the parabola $y = x^2 - 2x$ and the line $y = 4x + k$, where k is a constant. The line is tangent to the parabola at the point P .

- (a) Write down an equation whose solutions give the x -coordinates of the points where the line meets the parabola. [2]

$$x^2 - 2x = 4x + k \quad (\text{M1})$$

$$x^2 - 6x - k = 0 \quad (\text{A1})$$

- (b) Simplify your equation to the form $x^2 + bx + c = 0$, stating b and c in terms of k . [2]

$$x^2 - 6x + (-k) = 0 \quad (\text{A1})$$

$$b = -6, \quad c = -k \quad (\text{A1})$$

- (c) For the line to be tangent, state the condition on the discriminant. [1]

$$\Delta = 0 \quad (\text{exactly one solution}) \quad (\text{A1})$$

- (d) Find the value of k . [3]

$$\Delta = (-6)^2 - 4(1)(-k) = 36 + 4k \quad (\text{M1})$$

$$36 + 4k = 0 \quad (\text{A1})$$

$$k = -9 \quad (\text{A1})$$

- (e) Find the coordinates of the point P . [2]

$$x^2 - 6x + 9 = 0 \Rightarrow (x - 3)^2 = 0 \quad (\text{M1})$$

$$x = 3, \quad y = 9 - 6 = 3$$

$$P = (3, 3) \quad (\text{A1})$$

2. Let $f(x) = -x^2 + 4x + 5$.

(a) Write $f(x)$ in the form $-(x - h)^2 + k$. [3]

$$f(x) = -(x^2 - 4x) + 5 \quad (\text{M1})$$

$$= -(x^2 - 4x + 4) + 4 + 5 \quad (\text{A1})$$

$$= -(x - 2)^2 + 9 \quad (\text{A1})$$

(b) Write down the coordinates of the vertex of the parabola. [1]

$$\text{Vertex} = (2, 9) \quad (\text{A1})$$

(c) Write down the equation of the axis of symmetry. [1]

$$x = 2 \quad (\text{A1})$$

(d) Find the x -intercepts of the graph of f . [2]

$$-(x - 2)^2 + 9 = 0 \quad (\text{M1})$$

$$(x - 2)^2 = 9 \Rightarrow x - 2 = \pm 3$$

$$x = -1 \text{ and } x = 5 \quad (\text{A1})$$

(e) Write down the y -intercept. [1]

$$f(0) = 0 + 0 + 5 = 5 \quad (\text{A1})$$

$$y\text{-intercept} = (0, 5)$$

3. The equation $2x^2 - 4x + p = 0$ has two distinct real roots, where $p \in \mathbb{R}$.

(a) Find the discriminant of the equation in terms of p . [2]

$$\Delta = (-4)^2 - 4(2)(p) \quad (\text{M1})$$

$$= 16 - 8p \quad (\text{A1})$$

(b) Find the values of p for which the equation has two distinct real roots. [2]

$$\Delta > 0 \quad (\text{M1})$$

$$16 - 8p > 0 \Rightarrow p < 2 \quad (\text{A1})$$

(c) Find the values of p for which the equation has no real roots. [1]

$$\Delta < 0 \Rightarrow p > 2 \quad (\text{A1})$$

(d) Find the value of p for which there is exactly one real root, and find that root. [3]

$$\Delta = 0 \Rightarrow 16 - 8p = 0 \Rightarrow p = 2 \quad (\text{A1})$$

$$2x^2 - 4x + 2 = 0 \quad (\text{M1})$$

$$x^2 - 2x + 1 = 0 \Rightarrow (x - 1)^2 = 0$$

$$x = 1 \quad (\text{A1})$$

4. The parabola $y = x^2 - 2x - 3$ and the line $y = x + 1$ are drawn on the same axes.

(a) Show that the x -coordinates of the points of intersection satisfy

$$x^2 - 3x - 4 = 0. \quad [2]$$

$$x^2 - 2x - 3 = x + 1 \quad (\text{M1})$$

$$x^2 - 3x - 4 = 0 \quad (\text{A1}) \quad (\text{AG})$$

(b) Factorise $x^2 - 3x - 4$.

[1]

$$(x - 4)(x + 1) \quad (\text{A1})$$

(c) Hence find the coordinates of the two points of intersection.

[3]

$$x = 4: \quad y = 4 + 1 = 5 \quad \Rightarrow (4, 5) \quad (\text{A1})$$

$$x = -1: \quad y = -1 + 1 = 0 \quad \Rightarrow (-1, 0) \quad (\text{A1})(\text{A1})$$

(d) Find the equation of the axis of symmetry of the parabola.

[1]

$$x = \frac{-(-2)}{2(1)} = 1 \quad (\text{A1})$$

(e) Find the coordinates of the vertex of the parabola.

[2]

$$f(1) = 1 - 2 - 3 = -4 \quad (\text{M1})$$

$$\text{Vertex} = (1, -4) \quad (\text{A1})$$